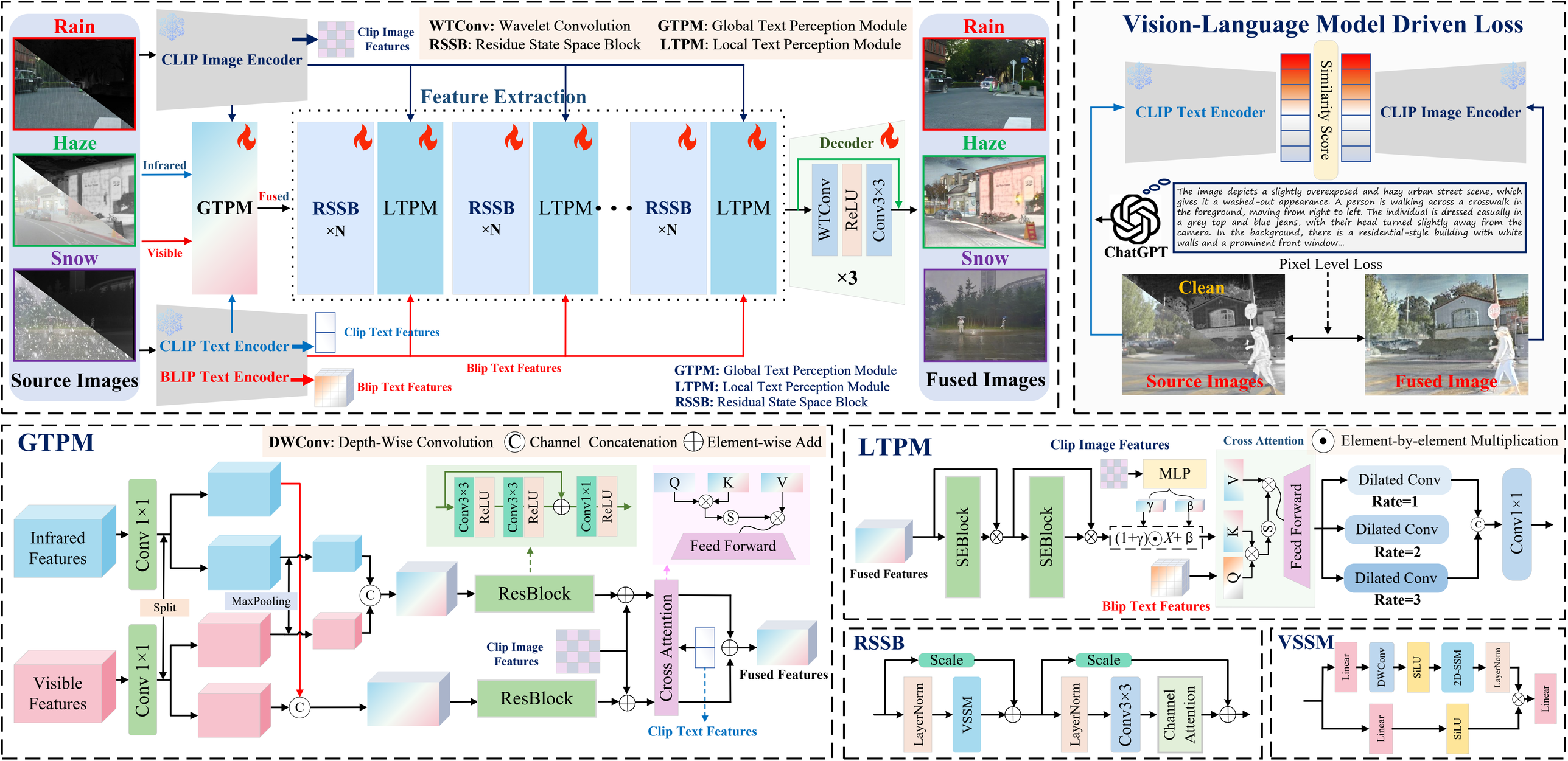
Rain
Haze
Snow
Source Images
CLIP Image Encoder
Clip Image Features
WTConv: Wavelet Convolution
RSSB: Residue State Space Block
GTPM: Global Text Perception Module
LTPM: Local Text Perception Module
Feature Extraction
Infrared
Visible
GTPM
Fused
RSSB ×N
LTPM
Decoder
WTConv
ReLU
Conv3×3
×3
CLIP Text Encoder
Clip Text Features
BLIP Text Encoder
Blip Text Features
GTPM: Global Text Perception Module
LTPM: Local Text Perception Module
RSSB: Residual State Space Block
Fused Images
Vision-Language Model Driven Loss
Similarity Score
ChatGPT
The image depicts a slightly overexposed and hazy urban street scene, which gives it a washed-out appearance. A person is walking across a crosswalk in the foreground, moving from right to left. The individual is dressed casually in a grey top and blue jeans, with their head turned slightly away from the camera. In the background, there is a residential-style building with white walls and a prominent front window...
Pixel Level Loss
Clean
Source Images
Fused Image
GTPM
DWConv: Depth-Wise Convolution
Channel Concatenation
Element-wise Add
Infrared Features
Visible Features
Conv 1×1
Split
MaxPooling
ResBlock
Clip Image Features
Cross Attention
Fused Features
Clip Text Features
Q
K
V
Feed Forward
LTPM
Clip Image Features
Cross Attention
Element-by-element Multiplication
Fused Features
SEBlock
MLP
γ
β
(1+γ)⊙X+β
Blip Text Features
Feed Forward
Dilated Conv
Rate=1
Rate=2
Rate=3
Conv1×1
RSSB
Scale
LayerNorm
VSSM
Conv3×3
Channel Attention
VSSM
Linear
DWConv
SiLU
2D-SSM
LayerNorm